One lecture

Slides 1…m

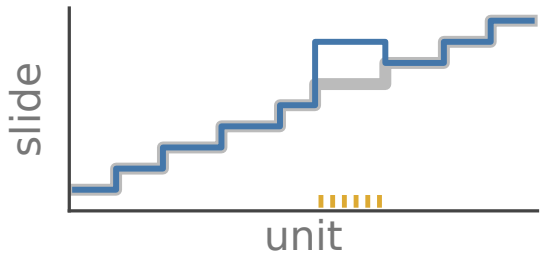
Units $u_1 \ldots u_n$
blue: important units

Similarity table
slides →
units ↓

8 measures: BM25, SPLADE, embeddings, cross-encoder, image, 3 hybrids
cost: tables (s), GPU

Method
31, in 8 families
Independent labelling
Monotone paths
Paths that may go back
Boundaries first
Transport
Consensus
LLM reads the text
LLM with similarity scores
cost: compute (s), 1 CPU core; LLM calls and LLM (s)

Cost per lecture = tables + compute + LLM, as if the method ran alone



One slide per unit
Accuracy = share of units on their true slide
Q4: important units on the wrong slide (% of all units)

The benchmark

4 public corpora
MaViLS, M³AV (A, B), MCD, Slide-Highlight

Preparation
1,131 lectures; ≤ 40 slides; units with a true slide

Attributes
7 attributes, 22 levels

Split
tuning
60 lectures
test
140 lectures
18,315 units

Tuning
31 methods × 8 measures; 25,773 combinations; on the 60 tuning lectures

Final run
31 methods, once, on the 140 test lectures

Results
Q1 ranking and cost
Q2 per level: length, pictures, tables
Q3 highest / lowest per level
Q4 important units on the wrong slide